Ref. [49] — Loss Landscape of Shallow ReLU-like Neural Networks

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1. The question and the obstacle

In a smooth objective, $\nabla L(\theta) = 0$ is the natural definition of stationarity. ReLU-like activations are non-differentiable at the origin, so their empirical squared-loss landscapes contain boundaries at which ordinary gradients do not exist. The authors ask which stationary points actually behave like minima, which admit an escape direction, and what happens when a narrow network is embedded inside a wider one. They analyze a *one-hidden-layer* model, not arbitrary deep networks.

2. First-principles mathematical setup

A representative scalar-output two-layer ReLU-like network has the form

$$f_\theta(x) = \sum_{j=1}^{m} a_j \sigma(w_j^\top x), \qquad L(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \big(f_\theta(x_i) - y_i\big)^2.$$

For ReLU, $\sigma(z) = \max\{0, z\}$. Away from a kink, differentiation through a unit gives

$$\frac{\partial L}{\partial w_j} = \frac{1}{n} \sum_{i=1}^{n} (f_\theta(x_i) - y_i) a_j \, \mathbf{1}\{w_j^\top x_i > 0\} x_i.$$

But at $w_j^\top x_i = 0$, the indicator can change discontinuously. The paper uses *directional stationarity* to work with these boundaries, rather than treating Clarke stationarity as interchangeable. The formula above is an explanatory smooth-region calculation, not the authors' full directional-stationarity definition.

3. What is an escape neuron?

A unit can have a parameter direction that would improve the squared loss even when the network is at a directional stationary point according to the paper's conditions. Such a unit is called an *escape neuron*; the authors characterize it with first-order criteria tailored to the activation's piecewise structure. Their central conclusions are logically asymmetric:

- At a stationary point *without* escape neurons, the point is a local minimum.
- For *scalar-output* networks, the presence of an escape neuron certifies that the stationary point is *not* a local minimum.

These results are not a claim that all local minima are global. Nor does identifying an escape neuron alone prove that plain gradient descent will escape within a particular iteration complexity.

> **Paper result.** Escape-neuron conditions distinguish saddle-like stationary configurations from local minima in one-hidden-layer ReLU-like models despite non-smooth activation boundaries.

4. Network embedding and saddle-to-saddle training

Embedding takes a narrower network and represents the same function with a wider architecture. For example, in a positively homogeneous model, splitting one neuron into two equal-output copies can preserve the prediction:

$$a\sigma(w^\top x) = \frac{a}{2}\sigma(w^\top x) + \frac{a}{2}\sigma(w^\top x).$$

This algebra is a teaching example; the paper develops a more complete account of how stationary configurations behave under embedding. It also narrows the types of saddle escape compatible with training from *vanishing initialization*. Its achievement is a more precise description of possible training paths, not a general guarantee of global convergence.

5. Assumptions, limitations, relevance

The model has one hidden layer, ReLU-like activations and empirical squared loss; the scalar-output qualifier matters for one direction of the escape-neuron result. The analysis does not cover arbitrary Transformer

objectives or provide a universal complexity rate for escaping non-smooth saddles. This is important for **RQ1** (critical-point geometry), **RQ2** (saddle-to-saddle dynamics), and **RQ4** (the gap between smooth theory and ReLU networks).

> **Do not conflate:** directional stationary points, Clarke stationary points, strict Hessian saddles and escape-neuron configurations are different objects. A nonsmooth landscape may not even have a Hessian at the relevant point.

6. Our notes

The main thing here is that ReLU kinks force us to be more careful about what "stationary" means. The escape-neuron idea gives a concrete way to tell whether a shallow network has a way to improve its loss, and embedding explains why the same learned function can look different in a wider network. It helps us discuss real activation functions without pretending the smooth strict-saddle theory applies directly. **Original source:** https://openreview.net/pdf?id=ogKE7LcvW6. **Verification:** original ICLR 2025 PDF and abstract checked; examples are marked as teaching derivations.